

ARJUN ADUKATHIL

College Station, TX | (979) 575-9041 | arjun.adukathil@tamu.edu | [LinkedIn](#) | [GitHub](#) | [Portfolio](#)

EXPERIENCE

LTIMindtree - ServiceNow

Jun 2021 – Jul 2024

Data Engineer

Mumbai, India

- Developed **150+ dbt models** across staging, ETL mart, and mart layers, applying **SCD patterns** and fact table granularity decisions to accurately represent business processes for 50+ stakeholders.
- Reduced backlog fact table ingestion **latency by 30%** by partitioning an **Airflow** pipeline by business unit, parallelizing DAG execution, and consolidating flows under a master DAG.
- Created a **config-driven Python data quality framework** that read validation rules from a Snowflake table, letting the ops team add or remove tests across 100+ datasets without modifying code, and automatically emailed alerts on failures.
- Saved **5+ engineering hours weekly** by automating Linux server storage cleanup with a **Bash cron job** applying file and directory retention rules across dev, test, and prod environments.
- Reduced dashboard load times **from 2+ minutes to under 30 seconds** by analyzing query plans, materializing views, applying **clustering**, and resolving expensive joins across Snowflake BI models.
- Migrated **20TB+** of data and all associated models from SAP HANA, SQL Server, and PostgreSQL to **Snowflake**, designing sales fact tables and dimensions in collaboration with the modeling team and data analysts.
- Engineered **Kafka** and **Spark** pipelines to ingest and transform **1M+ transaction records/hour** into S3, including end-of-day aggregation jobs for downstream consumption.
- Crafted **10+ KPI dashboards** in Tableau processing **2M+ records**, enabling executives to monitor sales performance, customer health, and product adoption trends.
- Reduced model failures and improved reusability by writing **dbt schema tests** covering null, unique, and referential integrity checks, and extracting repeated SQL logic into **macros**.

PROJECTS

Snowflake Data Ecosystem Analyst | *Cortex Agents, Streamlit, Python* | [GitHub](#)

Apr 2026 – May 2026

- Authored a semantic model mapping **6 Snowflake Account Usage views** to business concepts, enabling Cortex Analyst to translate plain-English questions into domain-accurate SQL with no manual query writing.
- Built a **3-tier fallback system** where Cortex Agents handles orchestration, a manual Chat Completions loop activates on failure, and regex-routed hardcoded SQL guarantees platform visibility during full AI outages.

SafePlate | *AWS Lambda, Amazon Bedrock, API Gateway, Flutter* | [GitHub](#)

Jan 2026 – Feb 2026

- Built a cross-platform Flutter app(iOS + Android) backed by **AWS Lambda + Amazon Bedrock** to extract menu items from photos and classify dishes into 3 allergen safety categories.
- Architected a **2-stage vision pipeline** using Claude 3.5 Sonnet, first extracting dishes via image analysis, then filtering by user allergen profile via a stateless REST API.

TerraLens | *PyTorch, NumPy, Scikit-learn, NetworkX, Jupyter* | [GitHub](#)

Oct 2025 – Dec 2025

- Achieved **56x faster similarity search** by building a scalable image-embedding pipeline using CNN features and HNSW indexing, reducing query latency from 37 ms to 0.66 ms.
- Improved clustering quality from **ARI 0.57 to 0.62** by constructing a 27k node k-NN graph and applying Louvain community detection to uncover high-purity data partitions.

New York City Trips Analytics | *GCP, GCS, BigQuery, DBT, Looker Studio* | [GitHub](#)

Jul 2025 – Aug 2025

- Built a scalable analytics platform using **BigQuery, DBT**, medallion architecture, and star-schema modeling that processed 10GB+ taxi data, enabling sub-second queries on trip pattern analysis.
- Supported data-driven decision-making by building interactive **Looker Studio dashboards** visualizing trip volumes, fare distributions, and geographic trends.

EDUCATION

Texas A&M University

Aug 2024 – May 2026

Master of Science in Management Information Systems, GPA: 3.75/4.0

College Station, TX

University of Mumbai

Aug 2017 – May 2021

Bachelor of Engineering in Computer Science, GPA: 3.76/4.0

Mumbai, India

TECHNICAL SKILLS

Programming Languages: Python, SQL, Dart, JavaScript, Bash, C/C++, Java

Data Engineering: Snowflake, DBT, Apache Airflow, Spark, SAP HANA, MongoDB, SQL Server, Hadoop, Oracle

Cloud & DevOps: AWS, GCP, Azure, Docker, Kubernetes, Jenkins, Git, Bitbucket, Linux, Jira

Analytics: Tableau, Power BI, Looker Studio, NumPy, Pandas, Matplotlib, Microsoft Excel

Certifications: Snowflake SnowPro Core, Professional Scrum Master I, IT Automation with Python by Google

Publication: Housing Society Management System using Flutter and Firebase, IRJET, March 2021